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CHAPTER 1

INTRODUCTION

Determining the physical location of a radio-frequency (RF) transmitter is a well-known problem commonly referred to as geolocation. Geolocation data can be used to find RF jammers or find unauthorized transmitters, locate RF tags, find or track people and vehicles, as well addressing many other problems [1–5]. The problem is inherently difficult for a few key reasons. Depending on the method used, a geolocation system may require expensive hardware in order to achieve high degrees of synchronization and coherency. Additionally, processing of large amounts of data can quickly become intractable for modern computers. Other methods that may not require expensive hardware may still be affected by electromagnetic phenomena such as multi-path distortion or fading [6, 7].

Among the most common methods for geolocation systems are time of arrival, time difference of arrival, angle of arrival, recieved signal strength indicators, doppler, pseudo-doppler, and the use of highly directional antennas. Of all of these methods, the one requiring the least amount of hardware generally is using received signal strength indicators (RSSI) [8].

The proposed research has three main objectives. First, to simulate a system where a stationary transmitter is located using multiple received signal strength measurements at different locations. Second, the research intends to examine the nonidealities of such a system, including non-quadratic power loss and sparse or clustered data, and to develop an algorithm to locate a transmitter in this non-ideal environment. Finally, the research will test this algorithm using real data obtained by using a software-defined radio paired with an Android phone running GNURadio. The novelty of this system and algorithm will be in its portability, commercial off-the-shelf availability, and relative robustness in terms of not requiring high degrees of synchronization or coherency.

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CHAPTER 2

Literature Review

2.1 Geolocation

Geolocation, as stated in the introduction, is useful for a number of reasons, and has a considerable amount of research addressing its complexities as well as different methods for obtaining accurate location estimates in a variety of settings. The most common methods for geolocation include [1, 9]:

- Time of Arrival (TOA)
- Time Difference of Arrival (TDOA)
- Angle of Arrival (AOA or sometimes DOA)
- Received Signal Strength Indicators (RSSI)

It is also interesting to note that geolocation using WiFi fingerprinting is commonly used on smart phones when GPS location services are unavailable. This method involves making databases of known locations associated with known signal strengths when GPS services are available, and saving these measurements for later use. This method generally has poor accuracy with comparison to other methods, unless further processing is done [10]. This problem is the inverse of the proposed research problem, where a receiver is trying to locate itself based on a known transmitter. For this reason, this method is not considered here.

2.2 TOA/TDOA/AOA

Time of arrival methods are sometimes referred to as time-of-flight methods, and can be used when the time of transmission is known. Besides requiring prior knowledge about the

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signal, this method suffers greatly from synchronization problems, when the clocks between receivers or between transmitters and receivers are not perfectly aligned [9, 11].

Time difference of arrival methods address the case where the exact transmission time of a signal is not known by the receivers. This is a commonly occurring case: searching for a signal without knowing exactly when it will be sent. By determining which receivers were able to capture a signal first, certain statements can be made about the location of the transmitter. Specifically, hyperbola can be drawn between receivers that represent a constant difference of arrival times between the two receivers. When multiple receivers are able to compare measurements, these hyperbola should intersect at the location of the transmitter. This method also suffers from synchronization problems like the TOA method [9]. Both of these time methods also suffer from not having direct line-of-sight, which is common in urban environments [7].

Angle of arrival methods rely on phase differences to determine the angle that the signal of interest is propagating from. Using an array of antennas (generally spaced at half-wavelength intervals) and assuming planar wave propagation, the phase difference between received signals at each antenna correspond to the physical direction of the transmitter. Perhaps even more than time-based methods, this phase method suffers greatly from multi path distortion [3, 7].

In general, these synchronization and multi path problems can be overcome using extensive processing, or improved hardware. The proposed research seeks to address this geolocation problem without the use of high-quality hardware or high degrees of time synchronization, so these methods will not suffice to solve the given problem. Instead, another method will be used (RSSI) which will serve both to simplify the receiver design, and make data collection less rigorous [12].

2.3 RSSI

Received signal strength indicators give a raw power measurement and can taken using relatively simple hardware [2, 4, 8, 13]. RSSI measurements are used to give measures of signal quality and roughly determine how far a receiver is from a transmitter [14]. In a

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perfectly ideal setting where the transmitted power is known, there is no noise, and the signal propagates through free space, the distance that the signal has traveled to reach the receiver can be perfectly known using Friis' transmission equation:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 R^2} \tag{2.1}$$

Of course, it is rarely the case that all of these ideal settings exist in a real system. Specifically, this research project seeks to determine the location of a transmitter where the power transmitted is not known, the signal is noisy, and the region between the receiver and transmitter is not free space.

Not knowing the transmitted power can be dealt with in a way similar to how TDOA methods operate without knowing transmission time. Measurements from multiple receivers can be used to draw circles (in free space) of constant power ratios between those receivers.

The problems associated with RSSI measurements are well known, and "the mathematical and statistical methods to tackle RSSI-variance problem needs further research." [6] The proposed research aims to do exactly that, address the non-idealities associated with using RSSI measurements for geolocation.

2.4 Geolocation using Robots

Distributed systems are well suited for taking RSSI measurements because the measurement is simple, and measurements must be taken at different physical locations. In practice, it can be practical to have one moving receiver take multiple points. For this reason, multiple studies have been done using drones or other autonomous robots to geolocate a transmitter [2, 3, 8, 9].

While the proposed research does not dictate measurements be taken autonomously through robots, it does propose the use of a mobile, distributed platform that allows a receiver to move physically and take multiple data points at different locations. For this reason, the work done in this area is applicable and provides useful insight into the possibilities available to a distributed network of portable receivers.

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CHAPTER 3

Research and Design Methods

3.1 Simulation Development

Evaluation of different algorithms can be done more easily in simulation than in practice. This will allow for the generation and testing of various types of data sets without the need to find and record large amounts of real-world data for every conceivable setting.

Simulated RSSI data will be generated for different settings, with different combinations of the following non-idealities:

- Unknown power transmitted
- Noise
- Non-Quadratic power loss
- Sparse data
- Clustered data

Generating data in these different cases will allow for rapid testing of algorithms without a need for physically obtaining many different types of real data sets.

3.2 Algorithms

The proposed research will develop algorithms for many of the easier cases in order to validate our data sets and simulation methods. The primary objective after making a robust data-generation method will be to develop algorithms resistant to the listed non-idealities. Of most concern is the non-quadratic power loss which comes from removing the free-space assumption, and the problems associated with sparse or clustered data sets.

Work has been done to fuse RSSI data with other data sets in order to overcome these problems [4,10]. However, the primary focus of the proposed research will be to develop the

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statistical models and algorithms that can overcome these problems as far as is reasonable with just the given RSSI data. It may be the case that the developed algorithms will not provide useful geolocation estimates in certain cases, and if that is true it will be the research's objective to justify and explain why this is the case.

3.3 Real Data Testing

In order to verify the performance of the algorithm, real-world data can be collected where the transmitter location is known. Practically, this requires two things: a known signal we can use to geolocate, and a simple receiver that can take RSSI measurements along with physical (GPS) coordinates.

The proposed research aims to use a mobile phone with a small software-defined radio (SDR) in order to record data. Using the phone's GPS, RSSI measurements can be location-stamped, and later offloaded to a computer for processing.

There are a few known signals that would be convenient for data collection. The simplest would be to use a local FM radio station whose location is known. Another possible signal is the narrow-band FM signal from a portable walkie-talkie. By having someone operate the walkie-talkie continuously while staying in a known location, data can be easily taken in almost any setting (direct line-of-sight, tree cover, urban etc.). By collecting data in various ways, the research will be able to test the developed algorithms in a variety of non-ideal settings.

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CHAPTER 4

Conclusion

Geolocation using RSSI measurements can help to solve common geolocation problems when high degrees of synchronization are not available or not practical. In addition, geolocation based off of RSSI measurements can combine well with networks of distributed receivers working together to locate a source since there is no time-dependent feature of the data.

By simulating RSSI measurements, algorithms can be developed that are well suited to addressing the problems associated with RSSI measurements, such as measurement noise, non-quadratic power loss, and sparse or clustered data. Simulation will help not only to develop algorithms, but to also identify which settings such a system could successfully perform geolocation in.

These algorithms can then be tested against multiple real-world measurement sets by recording RSSI data through the use of software-defined-radios attached to mobile phones. By careful design and statistical analysis, this research will explore the different possibilities for algorithms suited for this problem.

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